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## CASE STUDY

# Designing a Bank Management System Database

A Self-Driven SQL & Database Design Project

Built end-to-end in MySQL Workbench — 16 interconnected tables, 600+ realistic records, and a foundation for real-world SQL query practice.

*Prepared by a Data Analytics learner | MySQL Workbench 8.0 | 2026*

# Why I Built This Project

I recently completed a Data Analyst course, and one thing I kept hearing was: *SQL is best learned by working with real, messy, connected data — not just isolated practice tables*. So instead of jumping straight into solving random SQL problems, I decided to first build my own database from scratch — one that mimics how a real bank's backend data might actually be structured.

The goal of this phase was **not** to write complex queries yet — it was to design a realistic relational schema, understand primary and foreign key relationships, apply constraints correctly, and populate it with believable sample data. This project was built entirely in **MySQL Workbench**, table by table, relationship by relationship.

In the next phase, I'll be using this same database to practice real-world analytical SQL queries — joins across multiple tables, aggregations, subqueries, window functions, and building reports similar to what a bank's analytics team might actually need.

## Project Snapshot

|                          |                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------|
| Tool Used                | MySQL Workbench 8.0                                                                        |
| Database Name            | bank_db                                                                                    |
| Total Tables             | 16 interconnected tables                                                                   |
| Approx. Records Inserted | 600+ rows across all tables                                                                |
| Core Concepts Applied    | PK/FK relationships, ENUM, UNIQUE, DECIMAL precision, date handling, referential integrity |
| Purpose                  | Schema design practice → foundation for advanced SQL query practice                        |

# Schema Design Overview

The database, **bank\_db**, was designed to reflect how a real banking system separates its data — customer identity, branch network, staff records, financial products, transactional activity, and risk/compliance data are all kept in their own tables but connected through foreign keys. Here's a quick look at all 16 tables and what each one represents:

|                           |                                                                                                         |
|---------------------------|---------------------------------------------------------------------------------------------------------|
| <b>customers</b>          | Master record of every bank customer — KYC details, contact info, occupation, income, and risk profile. |
| <b>branches</b>           | All physical bank branches with IFSC codes, location, and status.                                       |
| <b>employees</b>          | Staff records per branch including designation, department, and reporting manager.                      |
| <b>accounts</b>           | Savings/current/salary accounts linked to customers and branches, with balance and nominee info.        |
| <b>cards</b>              | Debit/credit cards issued against accounts, with network, limits, and status.                           |
| <b>transactions</b>       | Day-to-day debit/credit activity — UPI, NEFT, RTGS, ATM, IMPS — with merchant and location data.        |
| <b>loans</b>              | Personal, home, car, and gold loans with tenure, EMI, and approval details.                             |
| <b>loan_payments</b>      | EMI payment history against each loan, including late fees and remaining balance.                       |
| <b>fixed_deposits</b>     | FD accounts with maturity tracking and interest rates.                                                  |
| <b>beneficiaries</b>      | Registered third-party payees for fund transfers.                                                       |
| <b>fund_transfers</b>     | NEFT/RTGS/UPI/IMPS transfers between accounts and beneficiaries.                                        |
| <b>insurance_policies</b> | Life, health, vehicle, home, and travel policies sold to customers.                                     |
| <b>credit_scores</b>      | Customer credit rating, utilization, and missed-payment history.                                        |
| <b>login_activity</b>     | Net-banking/app login logs — device, browser, IP, and failed attempts.                                  |
| <b>fraud_alerts</b>       | System-generated alerts for suspicious transactions and their resolution status.                        |

# Building It, Step by Step

Here's a walkthrough of the actual build inside MySQL Workbench — from creating the schema to populating each table with realistic sample data.

## 01 Database & Table Structure

The project starts with creating the **bank\_db** database and defining the **customers** table with proper constraints — **UNIQUE** on email, phone, Aadhaar, and PAN, **NOT NULL** on mandatory KYC fields, and an **ENUM** for marital status. The schema browser on the left confirms all 16 tables that make up the final design.

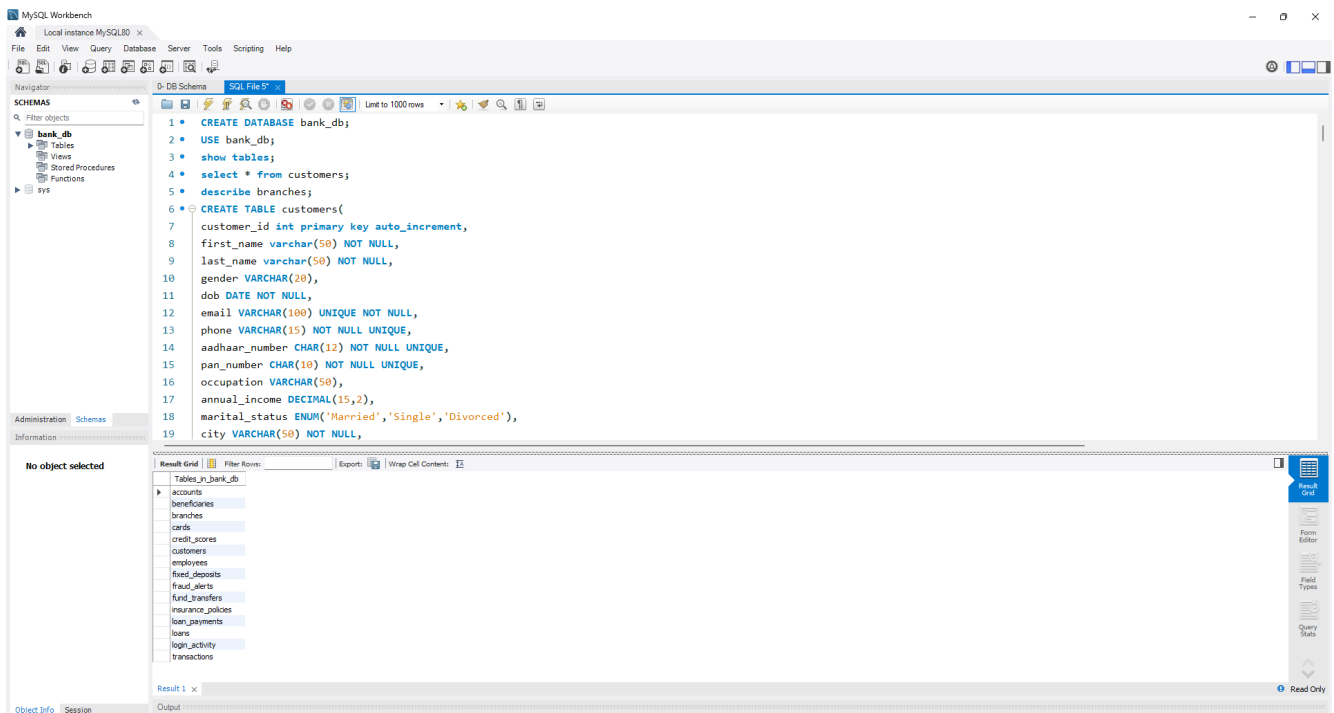


Fig 1 — Creating bank\_db and the customers table; full list of 16 tables in the schema.

# 02 Customer Master Data

25 customers were inserted with realistic Indian names, cities, occupations, income levels, and KYC status. Fields like **risk\_profile** and **customer\_since** were added to simulate what a real banking CRM would track.

| customer_id | first_name | last_name | gender | dob        | email                       | phone      | aadhaar_number | pan_number | occupation          | annual_income | marital_status | city       | state          | country | customer_since | kyc_status | risk_profile | created_at          |
|-------------|------------|-----------|--------|------------|-----------------------------|------------|----------------|------------|---------------------|---------------|----------------|------------|----------------|---------|----------------|------------|--------------|---------------------|
| 1           | Rahul      | Sharma    | Male   | 1988-05-12 | rahul.sharma1@gmail.com     | 9876500001 | 123456789001   | ABCDE1001F | Software Engineer   | 1200000.00    | Married        | Japur      | Rajasthan      | India   | 2020-01-15     | Active     | Low          | 2026-07-19 23:50:29 |
| 2           | Priya      | Verma     | Female | 1992-08-22 | priya.verma2@gmail.com      | 9876500002 | 123456789002   | ABCDE1002F | Teacher             | 650000.00     | Married        | Delhi      | Delhi          | India   | 2021-03-12     | Active     | Low          | 2026-07-19 23:50:29 |
| 3           | Amit       | Gupta     | Male   | 1985-11-08 | amit.gupta3@gmail.com       | 9876500003 | 123456789003   | ABCDE1003F | Business Owner      | 2500000.00    | Married        | Mumbai     | Maharashtra    | India   | 2019-07-01     | Active     | Medium       | 2026-07-19 23:50:29 |
| 4           | Neha       | Singh     | Female | 1995-02-18 | neha.singh4@gmail.com       | 9876500004 | 123456789004   | ABCDE1004F | Doctor              | 1800000.00    | Single         | Bangalore  | Karnataka      | India   | 2022-04-01     | Active     | Low          | 2026-07-19 23:50:29 |
| 5           | Vikas      | Yadav     | Male   | 1990-09-27 | vikas.yadav5@gmail.com      | 9876500005 | 123456789005   | ABCDE1005F | Sales Manager       | 900000.00     | Married        | Hyderabad  | Telangana      | India   | 2020-08-25     | Pending    | Medium       | 2026-07-19 23:50:29 |
| 6           | Pooja      | Jain      | Female | 1993-06-15 | pooja.jain6@gmail.com       | 9876500006 | 123456789006   | ABCDE1006F | CA                  | 1400000.00    | Single         | Ahmedabad  | Gujarat        | India   | 2021-11-18     | Active     | Low          | 2026-07-19 23:50:29 |
| 7           | Rohit      | Mehta     | Male   | 1987-12-09 | rohit.mehta7@gmail.com      | 9876500007 | 123456789007   | ABCDE1007F | Business Owner      | 3500000.00    | Married        | Chandigarh | Chandigarh     | India   | 2018-05-16     | Active     | High         | 2026-07-19 23:50:29 |
| 8           | Sneha      | Kapoor    | Female | 1996-04-30 | sneha.kapoor8@gmail.com     | 9876500008 | 123456789008   | ABCDE1008F | HR Manager          | 850000.00     | Single         | Lucknow    | Uttar Pradesh  | India   | 2022-01-09     | Active     | Low          | 2026-07-19 23:50:29 |
| 9           | Karan      | Mahotra   | Male   | 1991-10-14 | karan.mahotra9@gmail.com    | 9876500009 | 123456789009   | ABCDE1009F | Marketing Manager   | 1100000.00    | Married        | Indore     | Madhya Pradesh | India   | 2020-06-22     | Pending    | Medium       | 2026-07-19 23:50:29 |
| 10          | Anjali     | Saxena    | Female | 1989-01-20 | anjali.saxena10@gmail.com   | 9876500010 | 123456789010   | ABCDE1010F | Professor           | 1250000.00    | Married        | Pune       | Maharashtra    | India   | 2019-09-11     | Active     | Low          | 2026-07-19 23:50:29 |
| 11          | Sandeep    | Agarwal   | Male   | 1984-07-05 | sandeep.agarwal11@gmail.com | 9876500011 | 123456789011   | ABCDE1011F | Industrialist       | 4200000.00    | Married        | Japur      | Rajasthan      | India   | 2017-03-08     | Active     | High         | 2026-07-19 23:50:29 |
| 12          | Ritika     | Arora     | Female | 1997-05-18 | ritika.arora12@gmail.com    | 9876500012 | 123456789012   | ABCDE1012F | Software Engineer   | 950000.00     | Single         | Delhi      | Delhi          | India   | 2023-02-12     | Active     | Low          | 2026-07-19 23:50:29 |
| 13          | Manoj      | Triwari   | Male   | 1986-09-29 | manoj.triwari13@gmail.com   | 9876500013 | 123456789013   | ABCDE1013F | Bank Manager        | 1700000.00    | Married        | Mumbai     | Maharashtra    | India   | 2019-12-20     | Active     | Medium       | 2026-07-19 23:50:29 |
| 14          | Kavita     | Joshi     | Female | 1991-03-11 | kavita.joshi14@gmail.com    | 9876500014 | 123456789014   | ABCDE1014F | Architect           | 1300000.00    | Married        | Bangalore  | Karnataka      | India   | 2020-07-30     | Active     | Low          | 2026-07-19 23:50:29 |
| 15          | Nitin      | Bansal    | Male   | 1988-11-25 | nitin.bansal15@gmail.com    | 9876500015 | 123456789015   | ABCDE1015F | Trader              | 2800000.00    | Married        | Hyderabad  | Telangana      | India   | 2019-04-19     | Pending    | High         | 2026-07-19 23:50:29 |
| 16          | Meena      | Shah      | Female | 1994-09-08 | meena.shah16@gmail.com      | 9876500016 | 123456789016   | ABCDE1016F | Consultant          | 1150000.00    | Single         | Ahmedabad  | Gujarat        | India   | 2021-08-10     | Active     | Low          | 2026-07-19 23:50:29 |
| 17          | Arun       | Patel     | Male   | 1990-02-14 | arun.patel17@gmail.com      | 9876500017 | 123456789017   | ABCDE1017F | Civil Engineer      | 980000.00     | Married        | Chandigarh | Chandigarh     | India   | 2020-10-06     | Active     | Medium       | 2026-07-19 23:50:29 |
| 18          | Divya      | Gupta     | Female | 1995-12-27 | divya.gupta18@gmail.com     | 9876500018 | 123456789018   | ABCDE1018F | Data Analyst        | 850000.00     | Single         | Lucknow    | Uttar Pradesh  | India   | 2022-06-01     | Active     | Low          | 2026-07-19 23:50:29 |
| 19          | Harsh      | Mittal    | Male   | 1987-04-04 | harsh.mittal19@gmail.com    | 9876500019 | 123456789019   | ABCDE1019F | Business Owner      | 3200000.00    | Married        | Indore     | Madhya Pradesh | India   | 2018-08-15     | Active     | High         | 2026-07-19 23:50:29 |
| 20          | Payal      | Chopra    | Female | 1993-10-19 | payal.chopra20@gmail.com    | 9876500020 | 123456789020   | ABCDE1020F | Lawyer              | 1600000.00    | Married        | Pune       | Maharashtra    | India   | 2021-01-12     | Active     | Low          | 2026-07-19 23:50:29 |
| 21          | Deepak     | Kumar     | Male   | 1989-06-23 | deepak.kumar21@gmail.com    | 9876500021 | 123456789021   | ABCDE1021F | Government Officer  | 1050000.00    | Married        | Japur      | Rajasthan      | India   | 2019-05-24     | Active     | Low          | 2026-07-19 23:50:29 |
| 22          | Shweta     | Batra     | Female | 1996-07-17 | shweta.batra22@gmail.com    | 9876500022 | 123456789022   | ABCDE1022F | Teacher             | 620000.00     | Single         | Delhi      | Delhi          | India   | 2022-09-08     | Pending    | Low          | 2026-07-19 23:50:29 |
| 23          | Gaurav     | Rastogi   | Male   | 1989-01-09 | gaurav.rastogi23@gmail.com  | 9876500023 | 123456789023   | ABCDE1023F | Manufacturer        | 3900000.00    | Married        | Mumbai     | Maharashtra    | India   | 2017-11-13     | Active     | High         | 2026-07-19 23:50:29 |
| 24          | Komal      | Soni      | Female | 1994-03-21 | komal.soni24@gmail.com      | 9876500024 | 123456789024   | ABCDE1024F | HR Executive        | 780000.00     | Single         | Bangalore  | Karnataka      | India   | 2023-01-17     | Active     | Low          | 2026-07-19 23:50:29 |
| 25          | Yash       | Mehra     | Male   | 1992-12-05 | yash.mehra25@gmail.com      | 9876500025 | 123456789025   | ABCDE1025F | Marketing Executive | 720000.00     | Single         | Hyderabad  | Telangana      | India   | 2022-03-28     | Pending    | Medium       | 2026-07-19 23:50:29 |
| 26          | Aakash     | Pandey    | Male   | 1991-02-15 | aakash.pandey26@gmail.com   | 9876500026 | 123456789026   | ABCDE1026F | Software Engineer   | 1350000.00    | Married        | Ahmedabad  | Gujarat        | India   | 2020-04-18     | Active     | Low          | 2026-07-19 23:50:38 |

Fig 2 — Populated customers table with 25+ realistic records.

### 03 Branch Network

10 branches were added across major Indian cities, each with a unique IFSC code, branch type (Urban/Semi-Urban), and opening date — forming the base that employees and accounts link back to.

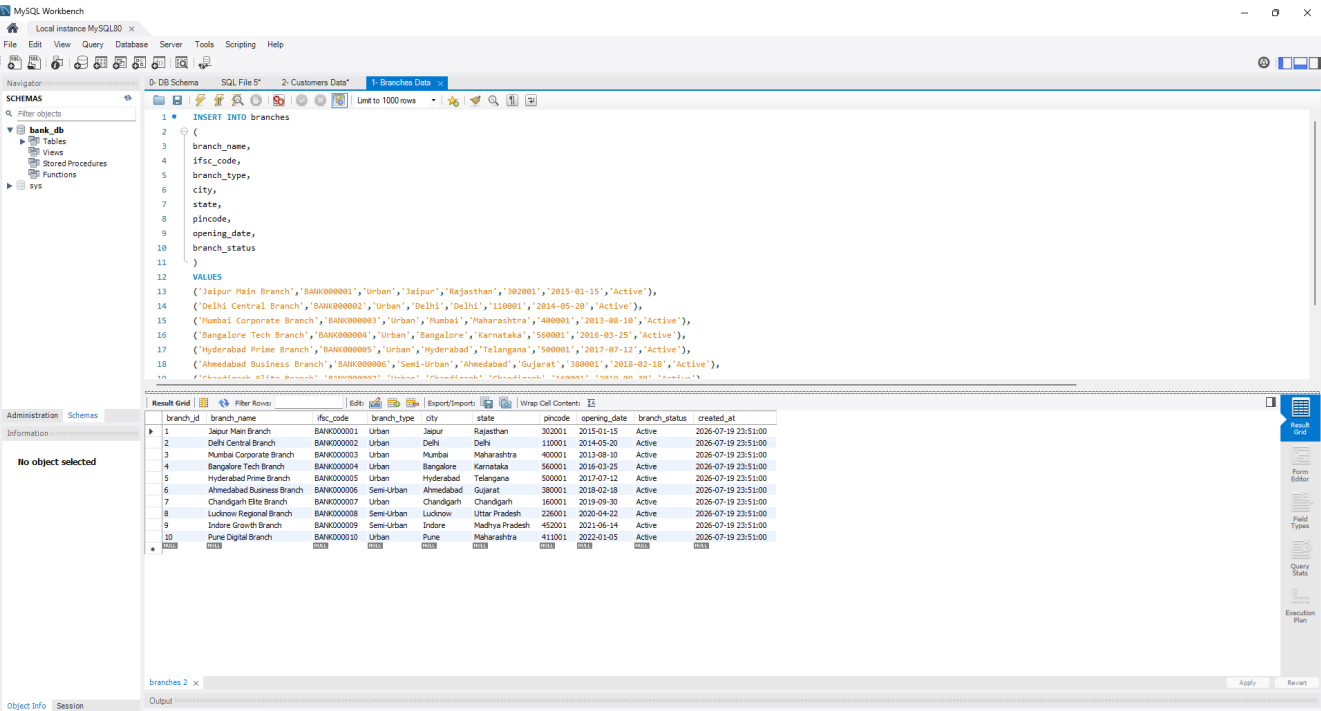
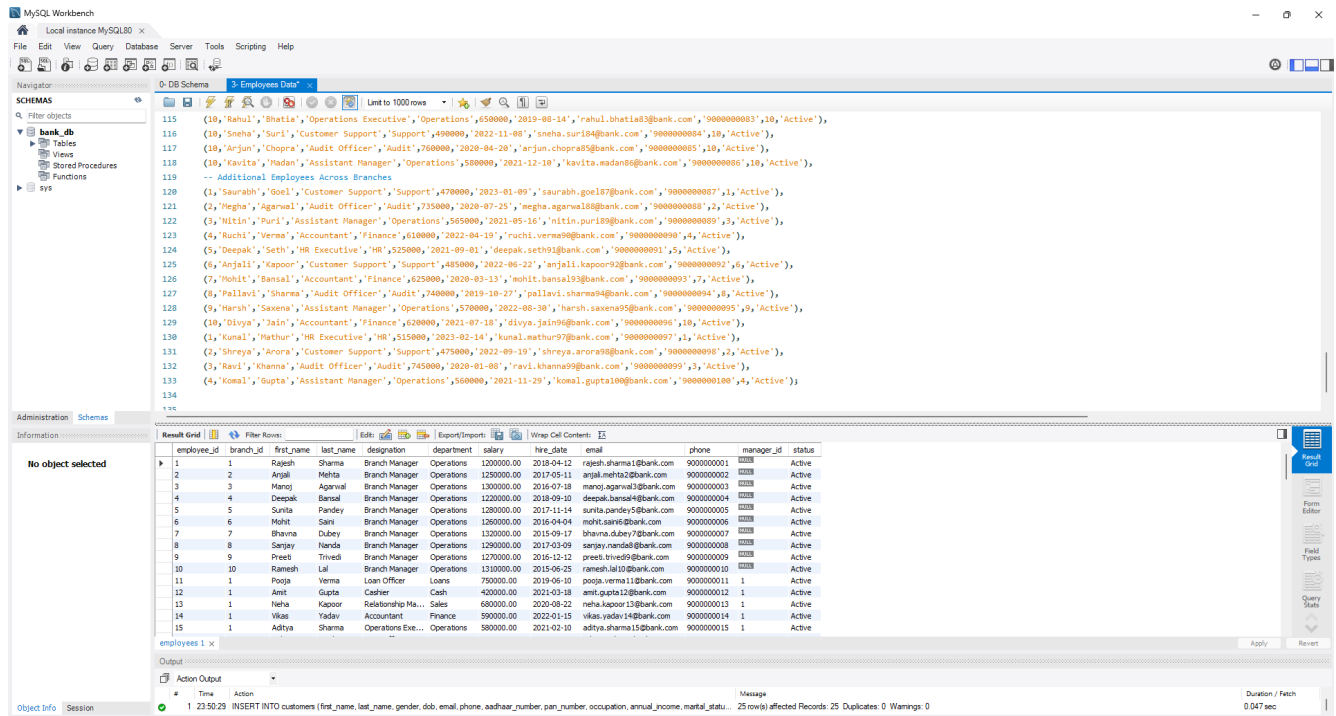


Fig 3 — Branch data with IFSC codes and location details.

## 04 Employee Records

Employees were mapped to branches with designations ranging from Branch Manager to Cashier, along with a self-referencing **manager\_id** to represent the reporting hierarchy — a good example of a self-join relationship for later query practice.



The screenshot displays the MySQL Workbench interface. The top pane shows the SQL editor with a query listing 130+ employee records. The bottom pane shows the result set of the query, which is a table with 13 columns: employee\_id, branch\_id, first\_name, last\_name, designation, department, salary, hire\_date, email, phone, manager\_id, and status. The table contains 130 rows of data, representing various employees across different branches and departments, including Branch Manager, Assistant Manager, Accountant, HR Executive, and Cashier. The manager\_id column indicates a self-referencing relationship for reporting hierarchy.

| employee_id | branch_id | first_name | last_name | designation        | department | salary    | hire_date  | email                    | phone      | manager_id | status |
|-------------|-----------|------------|-----------|--------------------|------------|-----------|------------|--------------------------|------------|------------|--------|
| 1           | 1         | Rajesh     | Sharma    | Branch Manager     | Operations | 120000.00 | 2018-04-12 | rajesh.sharma1@bank.com  | 9000000001 | 1          | Active |
| 2           | 2         | Arjun      | Mehta     | Branch Manager     | Operations | 125000.00 | 2017-05-11 | arjun.mehta2@bank.com    | 9000000002 | 1          | Active |
| 3           | 3         | Manoj      | Agarwal   | Branch Manager     | Operations | 130000.00 | 2016-07-18 | manoj.agarwal3@bank.com  | 9000000003 | 1          | Active |
| 4           | 4         | Deepak     | Bansal    | Branch Manager     | Operations | 122000.00 | 2018-09-10 | deepak.bansal4@bank.com  | 9000000004 | 1          | Active |
| 5           | 5         | Sunita     | Pandey    | Branch Manager     | Operations | 128000.00 | 2017-11-14 | sunita.pandey5@bank.com  | 9000000005 | 1          | Active |
| 6           | 6         | Mohit      | Sani      | Branch Manager     | Operations | 126000.00 | 2016-04-04 | mohit.sani6@bank.com     | 9000000006 | 1          | Active |
| 7           | 7         | Bhavna     | Dubey     | Branch Manager     | Operations | 122000.00 | 2015-09-17 | bhavna.dubey7@bank.com   | 9000000007 | 1          | Active |
| 8           | 8         | Sanjay     | Nanda     | Branch Manager     | Operations | 129000.00 | 2017-03-09 | sanjay.nanda8@bank.com   | 9000000008 | 1          | Active |
| 9           | 9         | Preeti     | Trivedi   | Branch Manager     | Operations | 127000.00 | 2016-12-12 | preeti.trivedi9@bank.com | 9000000009 | 1          | Active |
| 10          | 10        | Ramesh     | Lal       | Branch Manager     | Operations | 131000.00 | 2015-06-25 | ramesh.lal10@bank.com    | 9000000010 | 1          | Active |
| 11          | 1         | Pooja      | Verma     | Loan Officer       | Loans      | 75000.00  | 2019-06-10 | pooja.verma11@bank.com   | 9000000011 | 1          | Active |
| 12          | 1         | Amit       | Gupta     | Cashier            | Cash       | 42000.00  | 2021-03-18 | amit.gupta12@bank.com    | 9000000012 | 1          | Active |
| 13          | 1         | Neha       | Kapoor    | Relationship Ma... | Sales      | 68000.00  | 2020-08-22 | neha.kapoor13@bank.com   | 9000000013 | 1          | Active |
| 14          | 1         | Vikas      | Yadav     | Accountant         | Finance    | 59000.00  | 2022-01-15 | vikas.yadav14@bank.com   | 9000000014 | 1          | Active |
| 15          | 1         | Aditya     | Sharma    | Operations Exe...  | Operations | 58000.00  | 2021-02-10 | aditya.sharma15@bank.com | 9000000015 | 1          | Active |

Fig 4 — 130+ employee records across departments, with manager hierarchy.

# 05 Accounts

Each customer's accounts (Savings, Current, Salary) were linked via **customer\_id** and **branch\_id**, carrying balance, interest rate, and nominee details — the central table that most other financial activity connects back to.

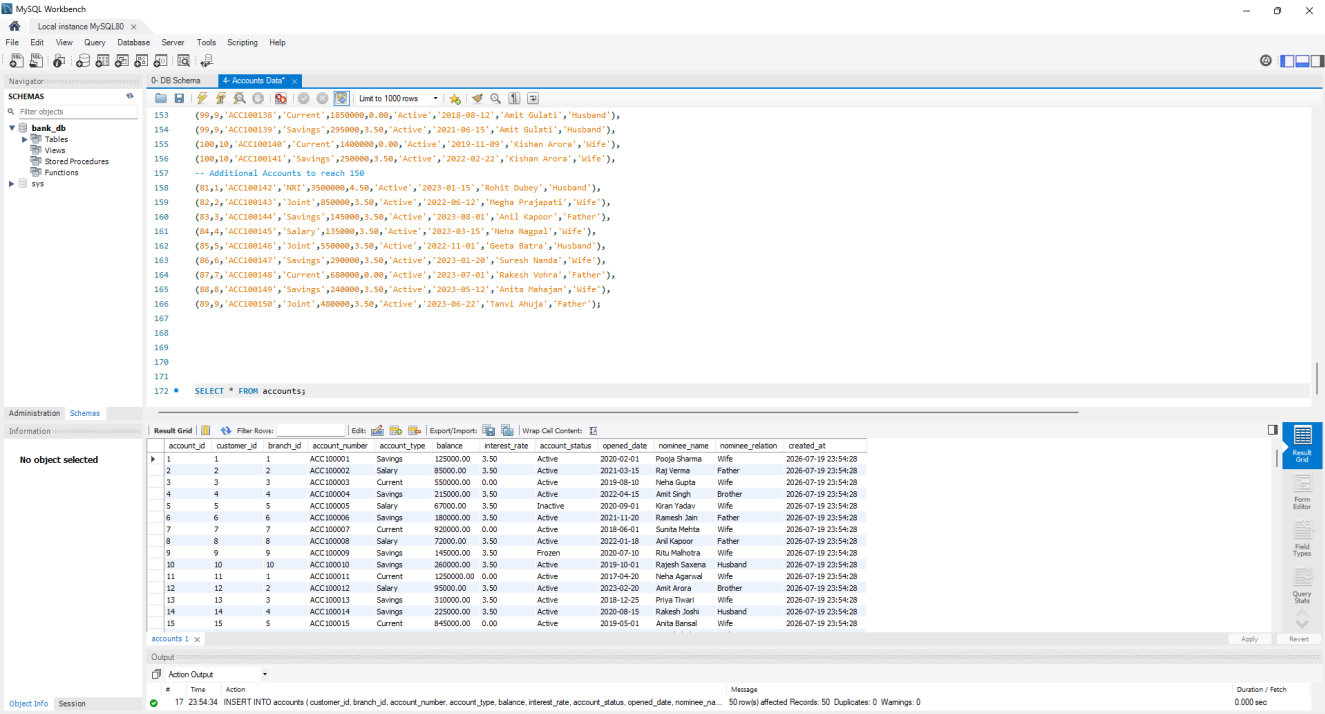


Fig 5 — 150 account records with balances and nominee information.



## 06 Cards

Debit and credit cards were issued against accounts, with card network (VISA/Mastercard/RuPay/AMEX), credit limits for credit cards only, and status tracking (Active/Blocked/Inactive).

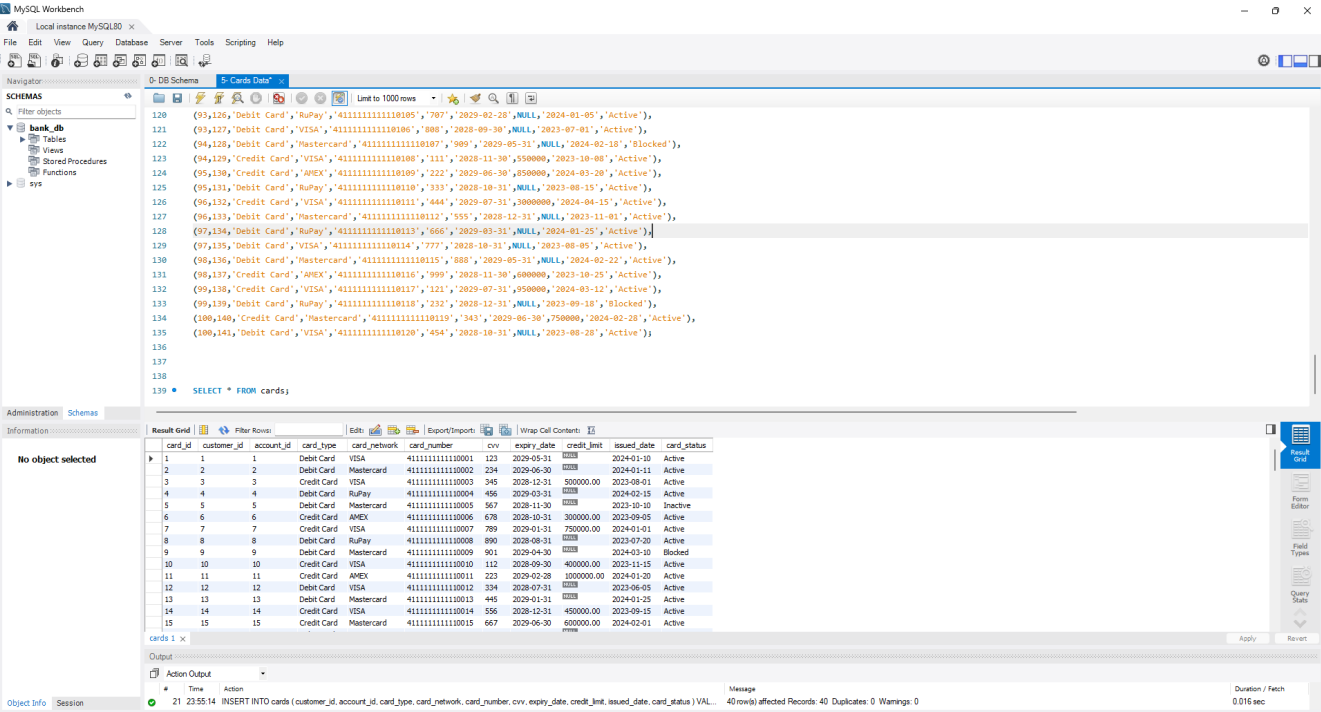


Fig 6 — Card issuance data tied to accounts and customers.

# 07 Transactions

Day-to-day transaction activity — UPI, NEFT, RTGS, ATM withdrawals — was generated with merchant names, categories, locations, and device/IP metadata, simulating what a transaction log table looks like in production systems.

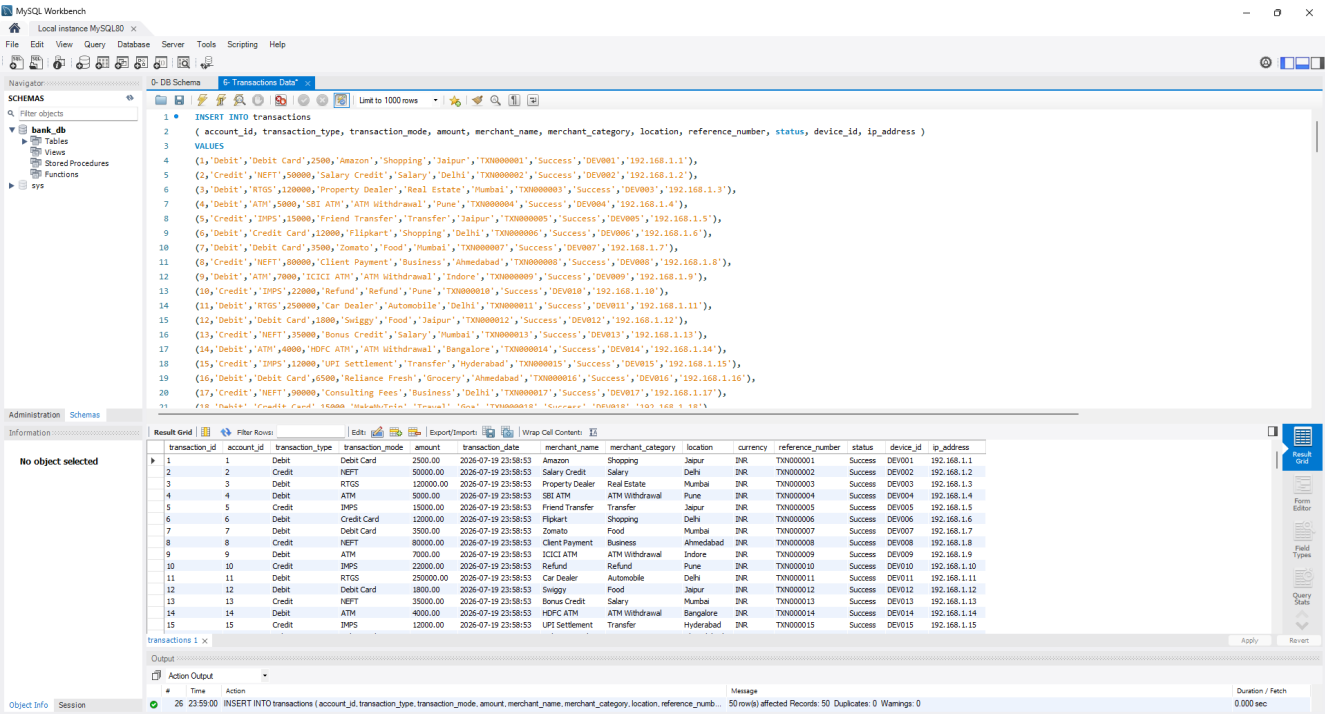


Fig 7 — Transaction log with merchant, category, and device metadata.

# 08 Loans

Personal, home, car, and gold loans were added with interest rate, tenure, EMI amount, and approval/disbursal dates, each tied to the approving employee — connecting the loans table back to employees.

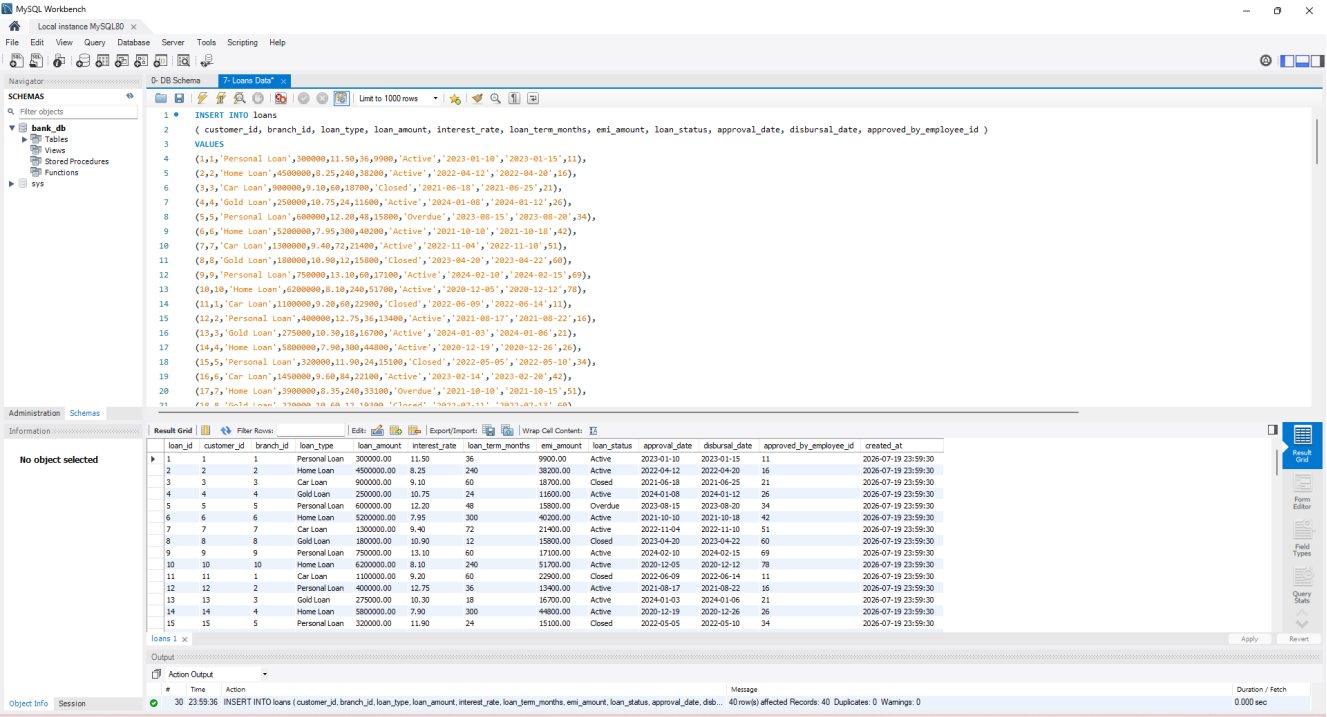


Fig 8 — Loan records with EMI, tenure, and approving officer.

# 09 Loan Payments

EMI payment history was tracked per loan, including late fees and running remaining balance — useful later for practicing running totals and window functions.

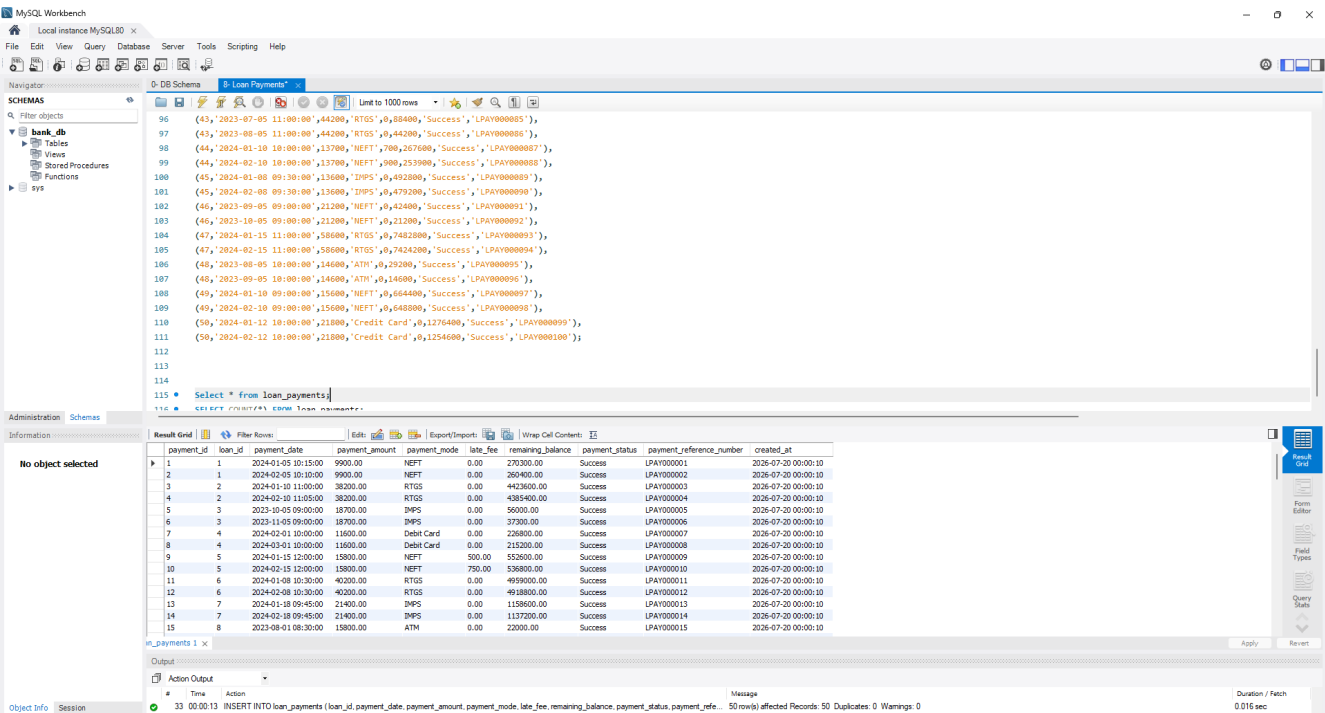


Fig 9 — EMI payment history with remaining balance tracking.

# 10 Fixed Deposits

FD records include principal, interest rate, start/maturity dates, and maturity amount — a good candidate for practicing date-difference and interest-calculation queries later.

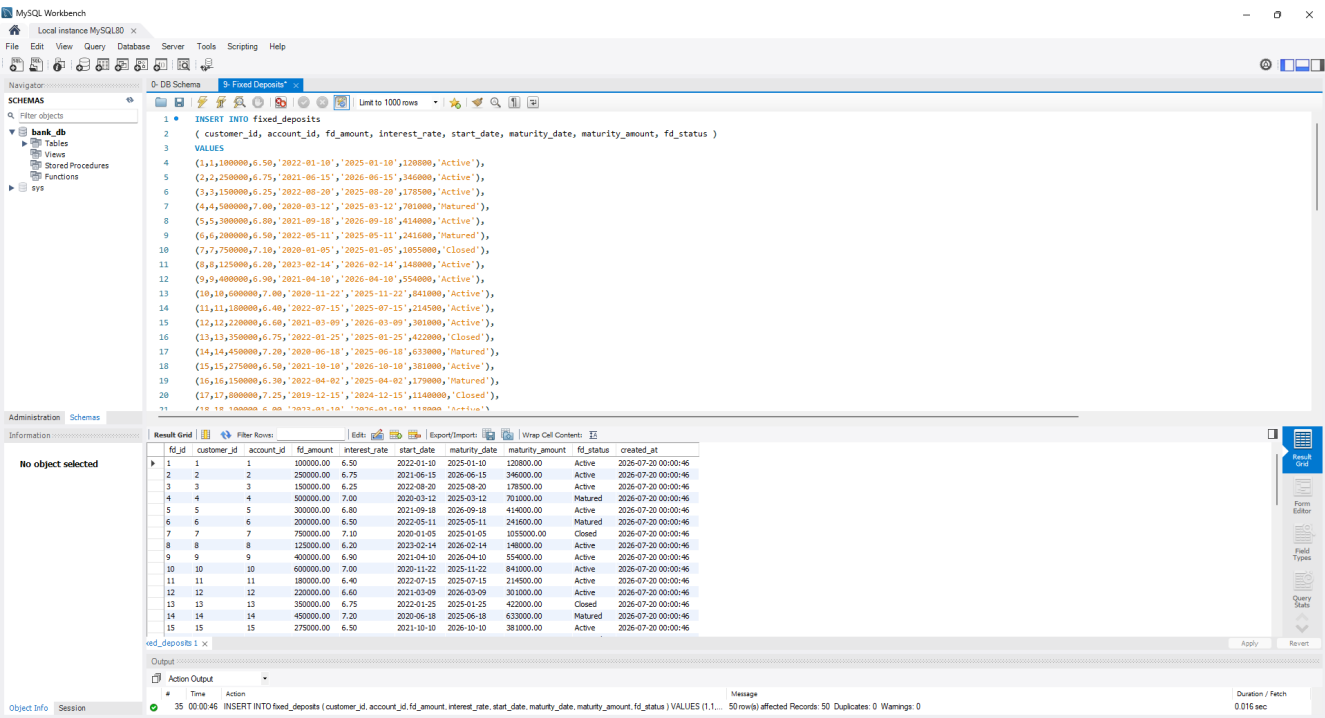


Fig 10 — Fixed deposit records with maturity tracking.

# 11 Beneficiaries

Registered payees were added per customer for fund transfers, each with bank name, IFSC, and relationship type — the link table between customers and the fund\_transfers table.

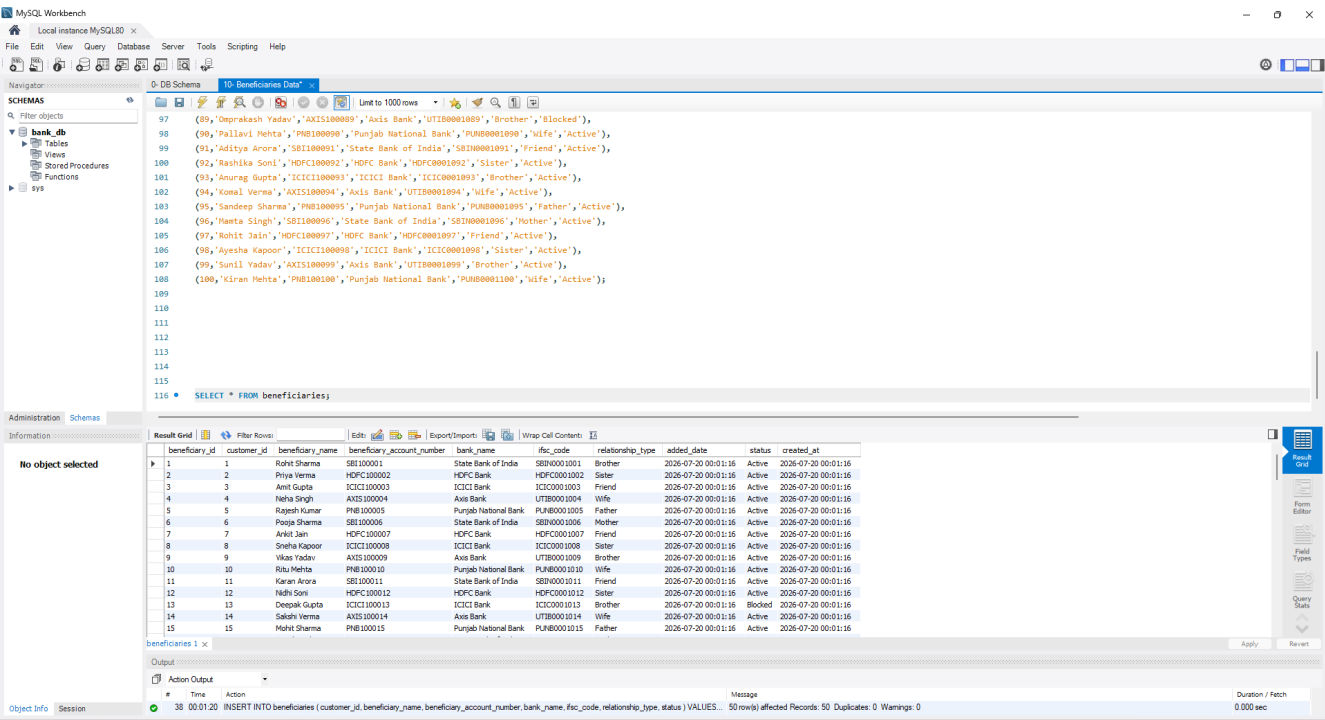


Fig 11 — Beneficiary records linked to customers.

## 12 Fund Transfers

Outward transfers (UPI/NEFT/RTGS/IMPS) between sender accounts and beneficiaries were logged with remarks and status — completing the money-movement side of the schema.

The screenshot displays the MySQL Workbench interface. The top pane shows an SQL query that inserts 15 rows into the `fund_transfers` table. The bottom pane shows the result grid with the following data:

| transfer_id | sender_account_id | beneficiary_id | transfer_type | amount    | transfer_date       | remarks             | transfer_status | reference_number | created_at          |
|-------------|-------------------|----------------|---------------|-----------|---------------------|---------------------|-----------------|------------------|---------------------|
| 1           | 1                 | 1              | UPI           | 2500.00   | 2026-07-20 00:01:42 | Monthly Support     | Success         | FT000001         | 2026-07-20 00:01:42 |
| 2           | 2                 | 2              | NEFT          | 15000.00  | 2026-07-20 00:01:42 | Family Transfer     | Success         | FT000002         | 2026-07-20 00:01:42 |
| 3           | 3                 | 3              | RTGS          | 250000.00 | 2026-07-20 00:01:42 | Property Payment    | Success         | FT000003         | 2026-07-20 00:01:42 |
| 4           | 4                 | 4              | IMPS          | 5000.00   | 2026-07-20 00:01:42 | Emergency Fund      | Success         | FT000004         | 2026-07-20 00:01:42 |
| 5           | 5                 | 5              | UPI           | 3500.00   | 2026-07-20 00:01:42 | Personal Transfer   | Success         | FT000005         | 2026-07-20 00:01:42 |
| 6           | 6                 | 6              | NEFT          | 22000.00  | 2026-07-20 00:01:42 | Household Expense   | Success         | FT000006         | 2026-07-20 00:01:42 |
| 7           | 7                 | 7              | IMPS          | 4500.00   | 2026-07-20 00:01:42 | Gift Transfer       | Success         | FT000007         | 2026-07-20 00:01:42 |
| 8           | 8                 | 8              | UPI           | 1800.00   | 2026-07-20 00:01:42 | Dinner Share        | Success         | FT000008         | 2026-07-20 00:01:42 |
| 9           | 9                 | 9              | RTGS          | 125000.00 | 2026-07-20 00:01:42 | Business Settlement | Success         | FT000009         | 2026-07-20 00:01:42 |
| 10          | 10                | 10             | NEFT          | 8000.00   | 2026-07-20 00:01:42 | Medical Expense     | Success         | FT000010         | 2026-07-20 00:01:42 |
| 11          | 11                | 11             | UPI           | 2200.00   | 2026-07-20 00:01:42 | Recharge Payment    | Success         | FT000011         | 2026-07-20 00:01:42 |
| 12          | 12                | 12             | IMPS          | 6000.00   | 2026-07-20 00:01:42 | Travel Expense      | Success         | FT000012         | 2026-07-20 00:01:42 |
| 13          | 13                | 13             | NEFT          | 25000.00  | 2026-07-20 00:01:42 | Family Support      | Success         | FT000013         | 2026-07-20 00:01:42 |
| 14          | 14                | 14             | UPI           | 3000.00   | 2026-07-20 00:01:42 | Monthly Allowance   | Success         | FT000014         | 2026-07-20 00:01:42 |
| 15          | 15                | 15             | RTGS          | 175000.00 | 2026-07-20 00:01:42 | Investment Transfer | Success         | FT000015         | 2026-07-20 00:01:42 |

The bottom status bar indicates: 42 00:01:48 INSERT INTO fund\_transfers (sender\_account\_id, beneficiary\_id, transfer\_type, amount, remarks, transfer\_status, reference\_number) VALUES (101,1... 50 row(s) affected Records: 50 Duplicates: 0 Warnings: 0

Fig 12 — Fund transfer log between accounts and beneficiaries.

# 13 Insurance Policies

Life, health, vehicle, home, and travel insurance policies sold to customers were added, with provider, premium, coverage amount, and policy status (Active/Expired/Cancelled).

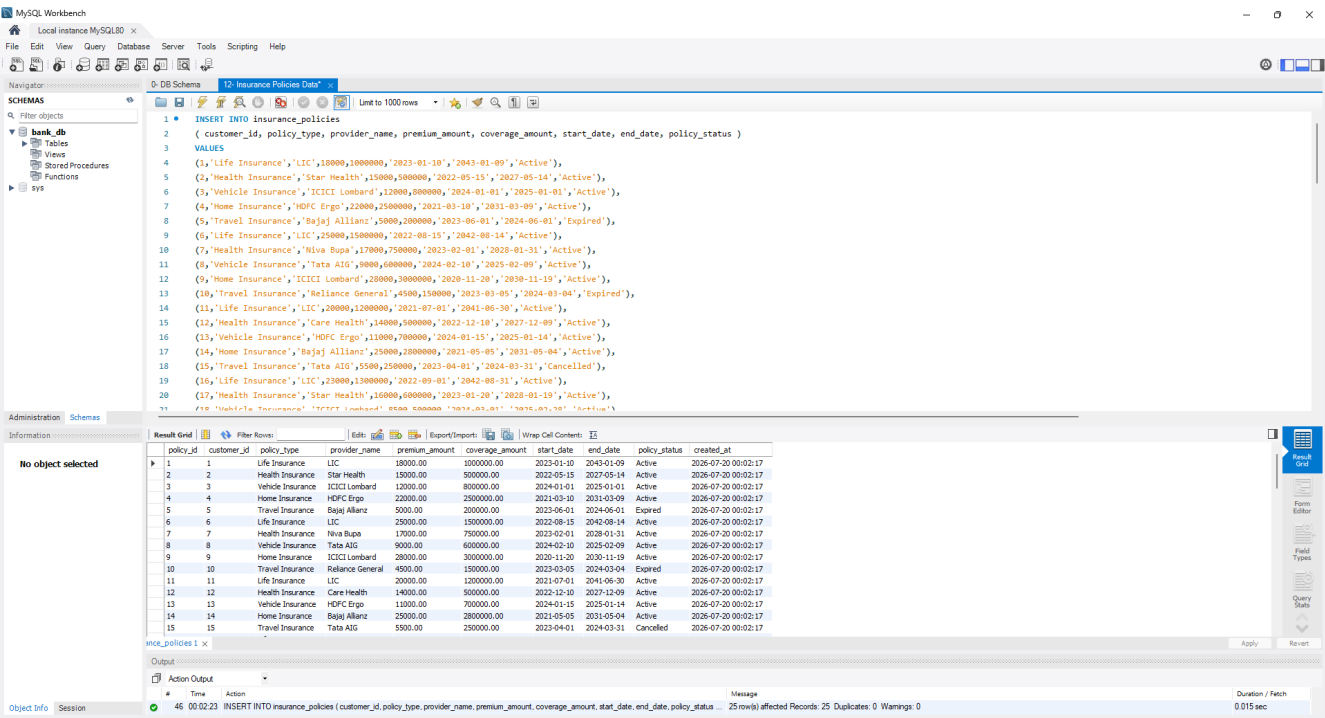


Fig 13 — Insurance policy data across multiple product types.



# 14 Credit Scores

A simplified credit-bureau-style table was added with score, rating, missed payments, and utilization percentage per customer — useful for later segmentation and risk-analysis queries.

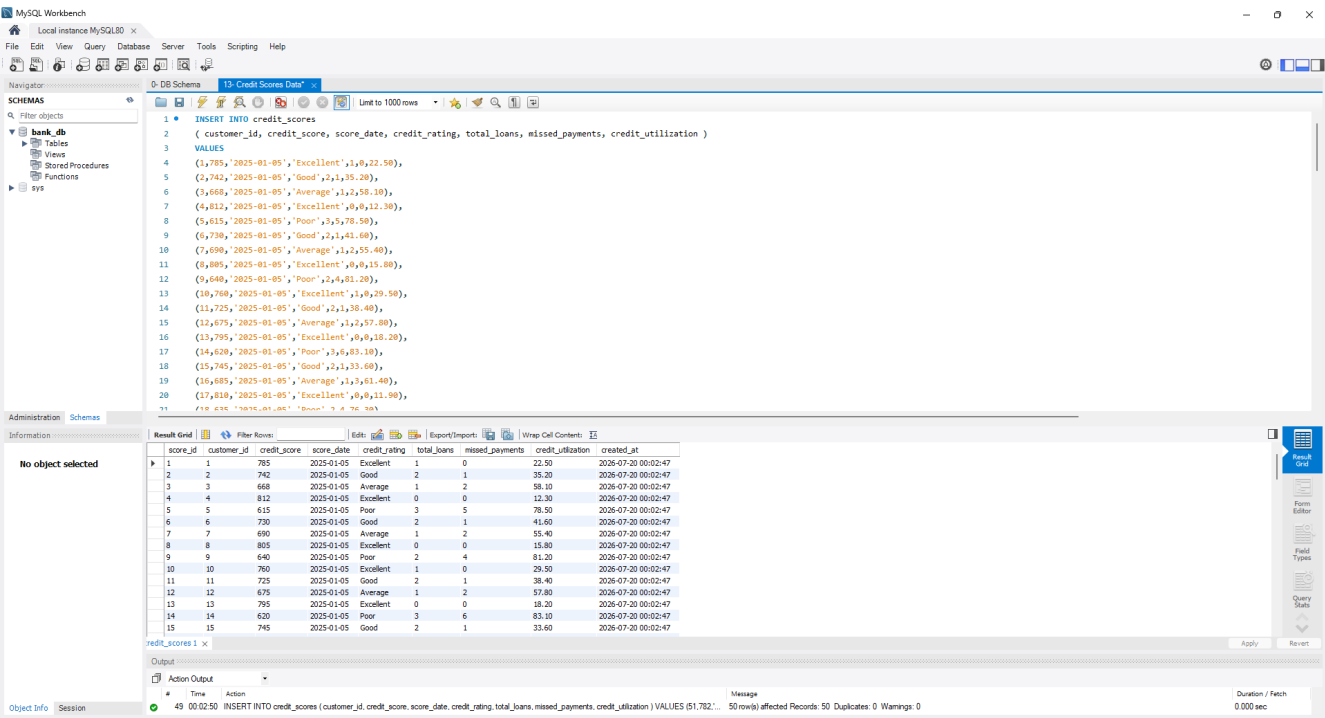


Fig 14 — Customer credit score and utilization data.

# 15 Login Activity

Net-banking/app login logs were simulated with device type, browser, IP address, failed attempts, and status (Success/Failed/Blocked) — a good dataset for later security/anomaly-detection style queries.

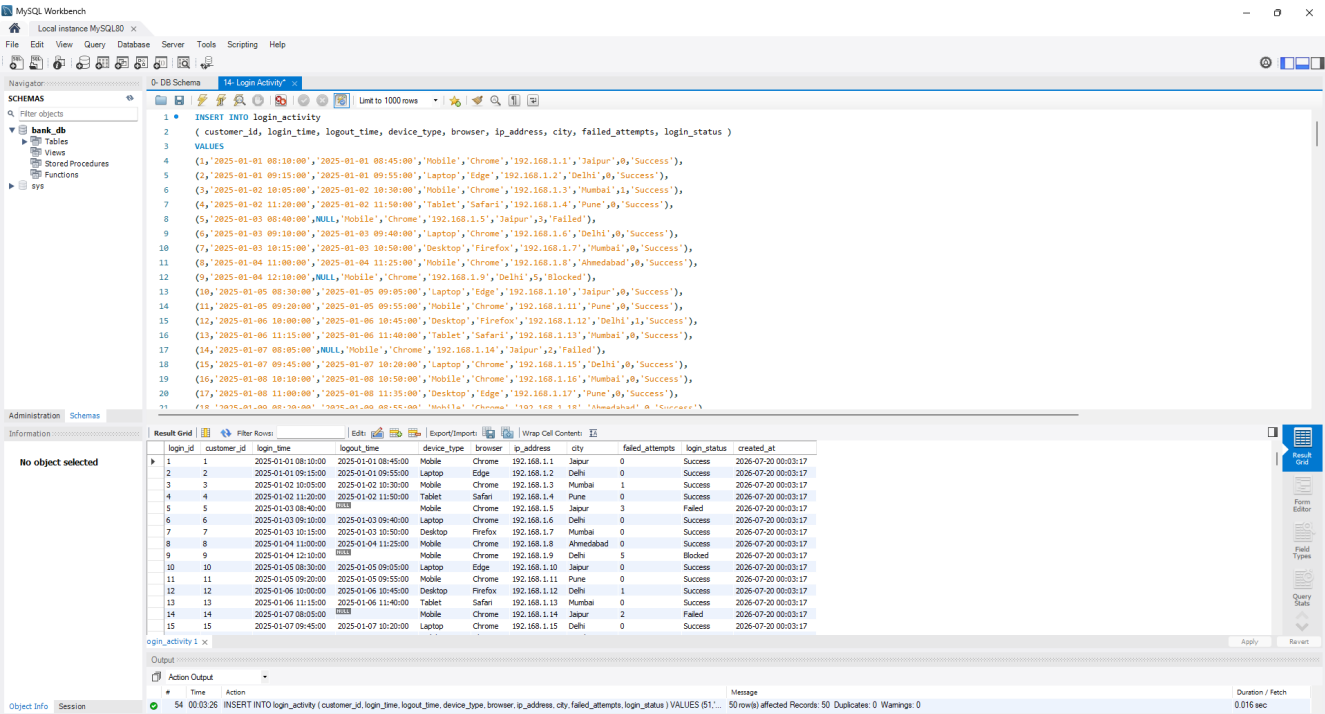


Fig 15 — Login activity log with device and status tracking.

# 16 Fraud Alerts

Finally, system-generated fraud alerts were added — high-value transactions, suspicious logins, foreign transactions — each with a risk score and resolution status, closing the loop on the risk-and-compliance side of the schema.

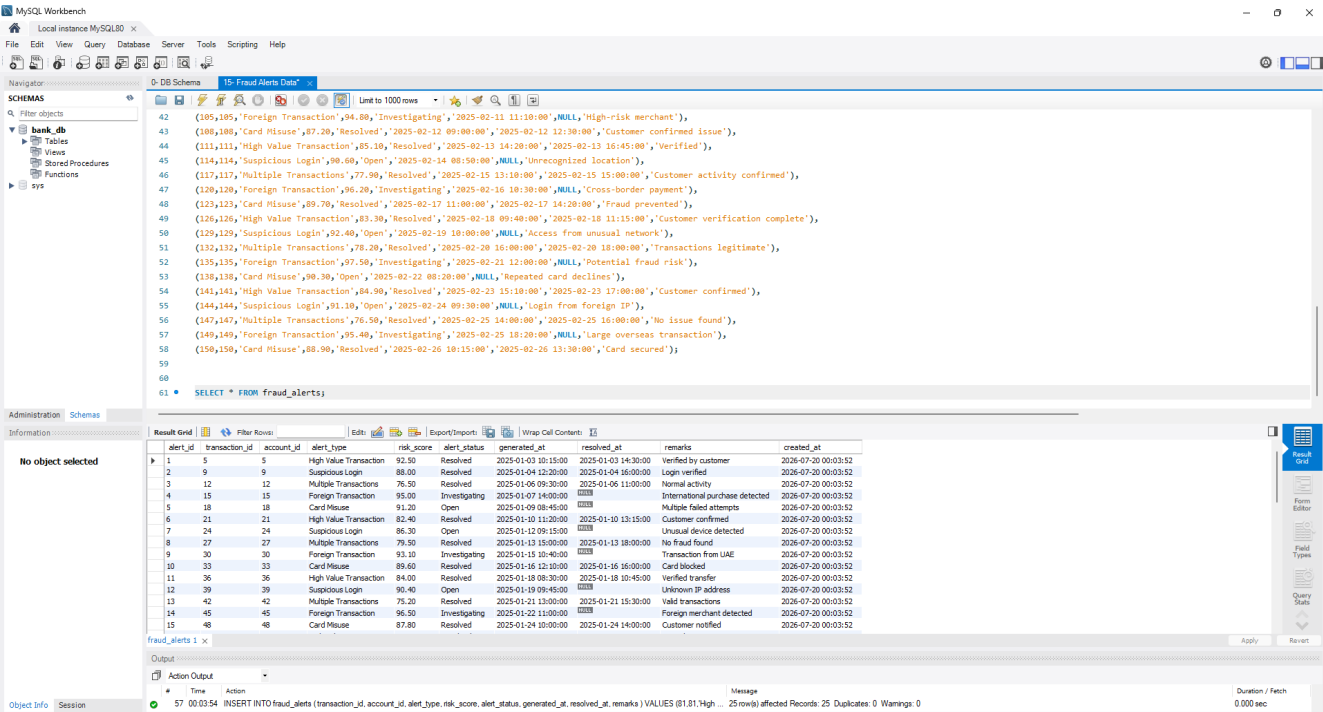


Fig 16 — Fraud alert records with risk score and resolution notes.

## What This Project Helped Me Practice

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- Designing a normalized, multi-table schema from a real-world domain instead of using pre-built datasets.
- Applying the right constraints — PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, ENUM, and CHECK-style logic — where they actually matter.
- Thinking through relationships: one-to-many (customers → accounts), self-referencing (employees → manager\_id), and many linking tables (accounts ↔ beneficiaries ↔ fund\_transfers).
- Writing realistic, internally consistent sample data (e.g., EMI amounts matching loan terms, remaining balances decreasing correctly across loan\_payments).
- Getting comfortable with MySQL Workbench itself — schema browser, result grids, bulk INSERT statements, and reading execution/output logs.

## What's Next

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With the schema and data in place, the next phase is where the real SQL practice begins — writing queries that a bank's data/analytics team might actually run:

- Multi-table JOINS — e.g., customer-wise total balance across accounts, FDs, and loans.
- Aggregations & GROUP BY — branch-wise transaction volume, average loan ticket size, NPA-style loan status breakdown.
- Window functions — running balance in loan\_payments, ranking customers by credit score within each city.
- Subqueries & CTEs — customers with more than one active loan, accounts with no fund transfers in the last 6 months.
- Fraud/risk analysis — flagging accounts with multiple fraud alerts or unusually high login failures.

This was a learning-first project, built purely to strengthen database fundamentals before diving into advanced SQL. If you've worked on something similar, or have suggestions on what real-world queries or additional tables I should add next — I'd genuinely love to hear your thoughts in the comments.